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An explainable hybrid transformer–BiLSTM framework for multivariate household energy demand forecasting with weather integration

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Abstract

Purpose Accurate and explainable household load forecasting is critical for demand-side management, tariff-aware scheduling, and reliable smart grid operation. This study introduces a leakage-controlled multi-horizon forecasting pipeline that integrates predictive accuracy with statistical validation, interpretability, robustness, and operational relevance.

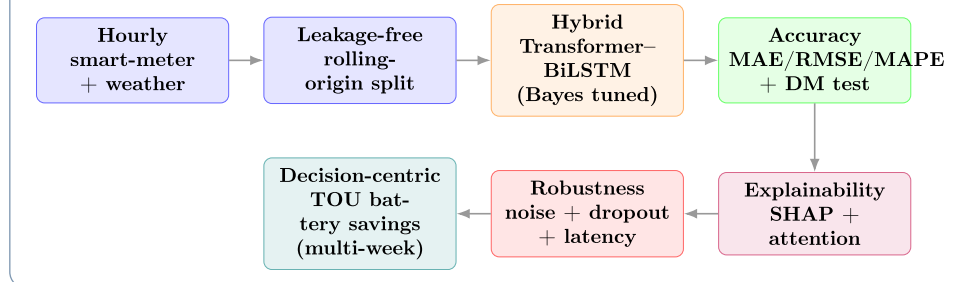
Methods We model multivariate household demand using an hourly smart-meter dataset spanning 14 months (Nov 2022–Jan 2024; $N=10,234$ time steps), incorporating aligned local weather covariates. A hybrid Transformer–BiLSTM is trained using a multi-output configuration to predict 24-hour and 168-hour trajectories. Hyperparameters are optimized via Bayesian optimization (Optuna) employing chronological train/validation/test splits and a rolling-origin evaluation protocol. Performance is assessed using MAE, RMSE, and MAPE, while pairwise forecast differences are validated using the Diebold–Mariano procedure. Model explanations are generated through SHAP and attention analyses, further complemented by robustness testing (noise and feature dropout) and inference-efficiency measurements.

Results At the 24-hour horizon, the hybrid model achieves MAE and RMSE values of 0.0539/0.0701 (MAPE 29.7%), yielding performance comparable to N-BEATS (MAE/RMSE 0.0531/0.0684). For the 168-hour horizon, the model attains superior performance among evaluated baselines (MAE/RMSE/MAPE 0.0566/0.0746/30.1%) and demonstrates statistically significant improvements over the Transformer, BiLSTM, and TCN models (Diebold–Mariano, $p < 0.001$). SHAP analysis identifies electrical indicators (e.g., voltage, power factor) and meteorological variables (e.g., pressure, temperature statistics) as the dominant drivers of medium-term predictions. Median inference latency remains in the tens-of-milliseconds range per sample, facilitating near real-time application.

Conclusion Beyond improving forecast accuracy, the proposed framework provides statistically supported and interpretable attributions, remains stable under input degradation, and demonstrates operational value in a tariff-based battery scheduling case study, reducing energy cost by approximately 2.29% over an aggregated multi-week evaluation.

Keywords Household energy demand forecasting, Smart meter data, Transformer–BiLSTM hybrid model, SHAP explainability, Bayesian optimization, Smart grid decision support

Graphical abstract



Introduction

Background and motivation

Accurate energy-demand forecasting is fundamental to the reliable and cost-effective operation of modern power systems. Given the difficulty of storing electricity at scale, supply must continuously match demand in near real-time; consequently, even moderate forecast errors can result in inefficient unit commitment, elevated balancing costs, and reliability risks [1, 2]. With rapid urbanization, electrification, and the proliferation of distributed energy resources, forecasting has become indispensable not only for short-term operations but also for maintenance planning, infrastructure investment, and policy formulation, facilitating sustainable development [3].

Within the context of smart grids, *short-term* horizons (e.g., the subsequent 24 hours) guide real-time dispatch, demand response, and dynamic pricing, while *medium-term* horizons (e.g., the upcoming 7 days or 168 hours) are critical for maintenance scheduling, hedging, and the allocation of flexibility resources [4]. Medium-horizon forecasting presents a particular challenge at the household level, as models must maintain stability amidst regime shifts caused by weekday/weekend routines, weather fronts, and behavioral variability. Despite these challenges, such forecasts are essential for multi-day energy management and tariff-based scheduling. Furthermore, reliable forecasts facilitate the integration of variable renewable energy sources by improving the alignment of flexibility resources and reducing avoidable CO₂ emissions [5].

At the household and building level, electricity loads exhibit strong seasonal cycles, calendar effects, and abrupt regime shifts driven by weather, occupancy, and behavioral variability, rendering the forecasting task highly nonlinear and stochastic [6]. Classical time-series models, such as AutoRegressive Integrated Moving Average (ARIMA) and exponential smoothing, are computationally efficient but often struggle to capture nonlinear multivariate dependencies and adapt to evolving data distributions [7, 8]. Similarly, while tree ensembles can capture feature interactions, they lack the inherent capacity to represent long-range sequential structures. Figure 1 depicts the residential energy ecosystem examined in this study, illustrating the data flow between smart meters, weather signals, and centralized analytics that enable forecasting and grid-aware decision-making.

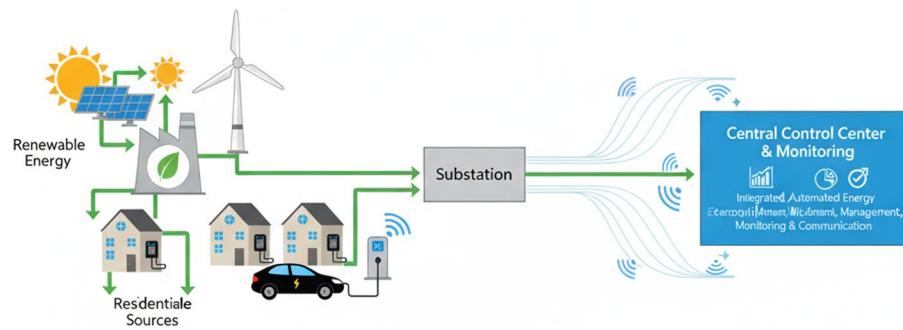


Fig. 1 Residential energy ecosystem connecting smart homes and distributed renewable resources to centralized control, forecasting, and decision-support systems

Related work

Classical and machine-learning baselines

Statistical forecasting remains widely used in operational settings due to transparency and speed. AutoRegressive Integrated Moving Average (ARIMA)/Error Trend, Seasonality (ETS) models can be competitive when dynamics are approximately linear and stationary, but performance degrades under multivariate, weather-driven, and cross-seasonal shifts typical of household demand [7, 8]. Machine-learning models and tree ensembles enhance nonlinear feature interactions, yet often require careful feature engineering and may struggle to capture multi-scale temporal dependencies without explicit sequence structure.

Deep learning for load forecasting

Deep learning (DL) architectures, including Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), Temporal Convolutional Networks (TCN), and Transformers, have shown strong performance in modeling complex temporal dependencies across demand scales [4]. LSTMs mitigate vanishing gradients through gated memory [9], while TCNs provide parallelizable receptive fields for temporal patterns. Transformers use self-attention to model long-range dependencies and seasonality, making them attractive for multi-horizon forecasting [10, 11].

Hybrid attention-recurrent architectures and multi-step strategies

Hybrid designs (e.g., CNN–LSTM, attention-augmented RNNs) often outperform single models by combining local pattern extraction with sequence memory [12]. Recent work has also combined Transformer-style attention with recurrent modules, reporting effective tracking of load fluctuations and decision-support use cases (e.g., building energy management systems) [13]. In parallel, multi-step-ahead forecasting strategy choice (direct versus rolling/recursive) has been shown to materially affect accuracy and efficiency across energy types and horizons [14].

Explainable forecasting frameworks for energy systems

Despite accuracy gains, many DL models remain “black boxes,” limiting interpretability, regulatory acceptance, and operational trust [15]. Explainable Artificial Intelligence (XAI) methods such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) provide post hoc attribution of model outputs to input features, while attention mechanisms can reveal salient temporal dependencies

[16]. Recent work has proposed explainable household forecasting frameworks by approximating Shapley values with DeepLIFT to map time-varying contributions of appliances and contextual factors [17]. Related studies in smart-building load forecasting integrate hybrid DL architectures with attribution methods (e.g., Integrated Gradients) to couple accuracy with transparency [18].

However, much of the explainability literature remains concentrated on short horizons and presentation-level interpretability, with limited linkage to operational stress regimes (e.g., heat-driven peaks, ramping periods) or downstream planning tasks. Moreover, many studies omit leakage-free temporal validation, formal statistical significance testing, robustness under realistic data degradation, and deployability considerations (Table 1).

Forecasting, renewables, and operational value

A recurring limitation in forecasting studies is the weak connection between error reductions and downstream energy value (e.g., cost savings under time-of-use tariffs, peak shaving, and flexibility scheduling) [19]. Beyond demand forecasting, the renewables forecasting literature increasingly emphasizes multi-energy integration, explainability, edge deployment, and uncertainty quantification to support operational decisions [20]. These directions motivate household-level frameworks that quantify value and reliability, not only accuracy.

Research gaps and contributions

Research gaps

The above review highlights several gaps that remain under-addressed in household energy demand forecasting:

- **Multi-horizon rigor:** limited studies jointly evaluate 24 h and 168 h horizons under strictly leakage-free temporal validation with formal statistical comparison.
- **Explainability-to-action:** Explainability is often post hoc and weakly connected to operational stress regimes (e.g., heat-driven peaks, ramp periods) and decision-support needs.
- **Reliability under data imperfections:** Robustness to noisy weather inputs and missing sensor channels is not consistently assessed.

Table 1 Checklist comparison of related forecasting studies

Feature	[17]	[21]	[13]	[14]	[20]	This work
Task (Load/Wind/PV)	Load	Load	Load	Load (multi-energy)	Wind	Load
Multi-horizon	✓	✓	✓	✓	✓	✓
Weather/exogenous inputs	✓	✓	✓	✓	✓	✓
Leakage-free temporal split	×	×	×	✓	–	✓
Statistical tests (e.g., DM)	×	×	×	×	–	✓
Explainability (SHAP/IG/...)	✓	✓	×	×	✓	✓
Robustness/sensitivity analysis	×	×	×	×	✓	✓
Complexity (Big-O)	×	×	×	×	–	✓
Latency/throughput reported	×	×	×	✓	✓	✓
Economic/planning value shown	✓	×	✓	✓	✓	✓

Legend: ✓ indicates the feature is explicitly reported/evaluated in the cited work; × indicates not reported; “–” denotes not applicable or not assessed (e.g., review papers). Abbreviations: DM, Diebold–Mariano test; XAI, explainable AI; IG, Integrated Gradients

- **Deployability evidence:** Computational complexity (Big-O), inference latency, and throughput are frequently omitted despite relevance to real-time controllers and edge deployment.
- **Economic impact quantification:** Forecasting accuracy improvements are seldom translated into clearly reported cost or operational benefits.

Contributions

To address these gaps, this paper makes the following contributions:

- We propose an end-to-end explainable forecasting framework based on a hybrid Transformer-BiLSTM architecture designed for multi-horizon (24 h, 168 h) household load forecasting with weather integration.
- We adopt a rigorous evaluation protocol featuring time-ordered rolling validation, Bayesian hyperparameter optimization, multi-metric assessment, and statistical significance testing using the Diebold-Mariano procedure.
- We provide physically interpretable explanations via SHAP and attention analyses and explicitly relate error regimes to temperature quantiles and ramp periods that are operationally relevant.
- We assess robustness and deployability through noisy/missing-input experiments and report inference latency/throughput and computational complexity, complemented by an illustrative tariff-driven battery scheduling case study that quantifies economic implications.

Figure 7 presents a conceptual map of the proposed end-to-end workflow, linking data acquisition and preprocessing, supervised windowing, model training and Bayesian hyperparameter optimization, evaluation, explainability, robustness analysis, and planning relevance.

Paper organization

The remainder of this paper is organized as follows. Section "[Materials and methods](#)" describes the dataset, preprocessing steps, and the proposed forecasting methodology. Section "[Results](#)" presents experimental results, including forecasting performance, statistical validation, interpretability, robustness, and efficiency analyses. Section "[Discussion](#)" discusses findings in the context of energy-system operation and planning. Section "[Conclusion](#)" concludes the paper and outlines directions for future research.

Materials and methods

Dataset and study setting

Data source and temporal scope

The study uses a household smart-meter dataset enriched with aligned meteorological covariates and resampled to an hourly resolution for multi-horizon forecasting. The hourly table contains $N = 10,234$ timestamps spanning from 2022-11-05 14:00:00 to 2024-01-05 23:00:00. The target variable is hourly active power consumption (`active_power`). Exogenous covariates include electrical measurements (`current`, `voltage`, `reactive_power`, `apparent_power`, `power_factor`) and weather variables (`temp`, `feels_like`, `temp_min`, `temp_max`, `pressure`, `humidity`, `speed`, `deg`).

Data quality checks and cleaning

We performed integrity checks on the hourly time index. The dataset contains no missing entries (total missing values = 0), no duplicate timestamps, and no missing hourly intervals across the covered period; therefore, no imputation was required. All transformations used for modelling (scaling and any time/calendar encodings) were fit using the training split only and then applied unchanged to validation and test subsets to prevent temporal leakage.

Feature set and dimensions at each processing step

From the full hourly table (shape 10,234 × 19), we retained the numeric feature set used for model training (13 covariates) and excluded any leakage-prone forward-looking variables (e.g., temp_{t+1}). The resulting matrices are

$$X_{\text{all}} \in \mathbb{R}^{10,234 \times 13}, \quad Y_{\text{all}} \in \mathbb{R}^{10,234}. \quad (1)$$

We use chronological splits: train (7, 163 rows), validation (1, 535 rows), and test (1, 536 rows), with split boundaries at 2023-08-31 00:00:00 (train/val) and 2023-11-02 23:00:00 (val/test). For supervised learning, a sliding window transforms the series into input-output pairs. For window length $w \in \{168, 336\}$ and horizon $h \in \{24, 168\}$, the windowed tensors have shapes:

$$X \in \mathbb{R}^{N^* \times w \times 13}, \quad Y \in \mathbb{R}^{N^* \times h}, \quad (2)$$

with $(N^*, w, h) \in \{(10043, 168, 24), (9899, 168, 168), (9875, 336, 24), (9731, 336, 168)\}$.

Exploratory data analysis and visualization

We present basic exploratory analysis to characterize the target dynamics, seasonal structure, and exogenous covariates. Figure 2 shows the hourly target series with chronological train/validation/test boundaries. To highlight systematic periodicity, Figs. 3 and 4 report the mean demand profile by hour-of-day and day-of-week, respectively. Figure 5 summarizes the distribution of the weather covariates using boxplots (outliers suppressed for readability). Finally, Fig. 6 shows the correlation structure among numeric variables, revealing strong collinearity among temperature-related fields (e.g.,

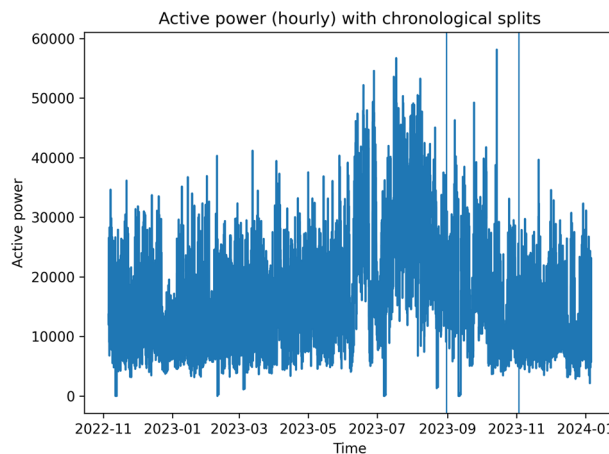


Fig. 2 Hourly active power series with chronological train/validation/test splits

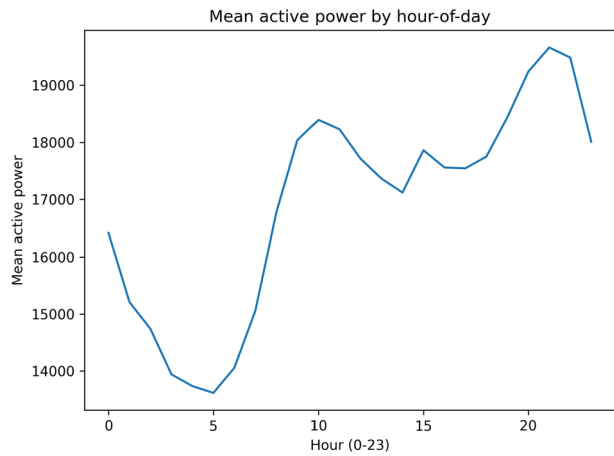


Fig. 3 Mean active power by hour-of-day (0–23)

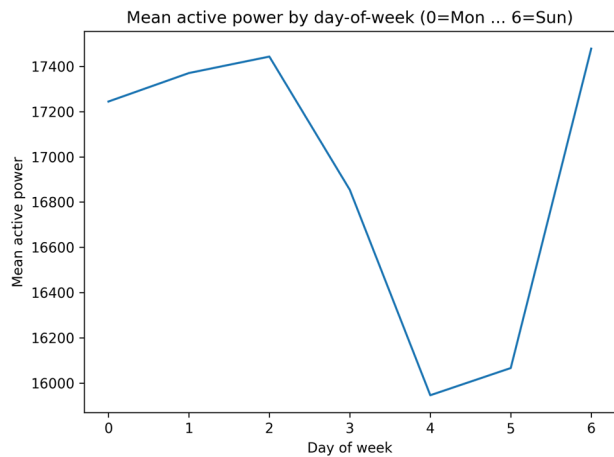


Fig. 4 Mean active power by day-of-week (0=Mon,...,6=Sun)

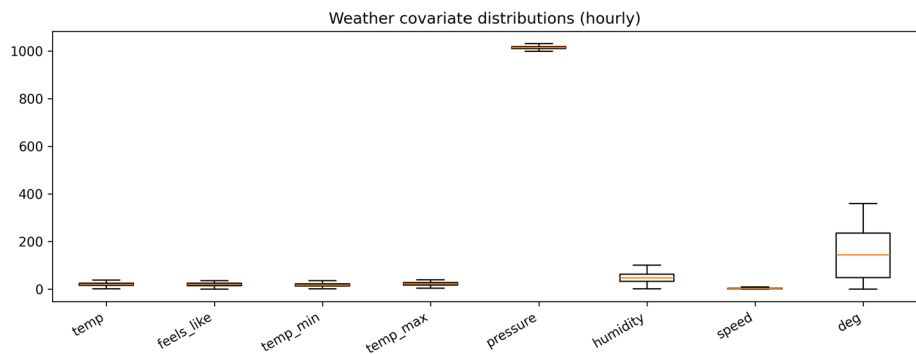


Fig. 5 Weather covariate distributions at hourly resolution

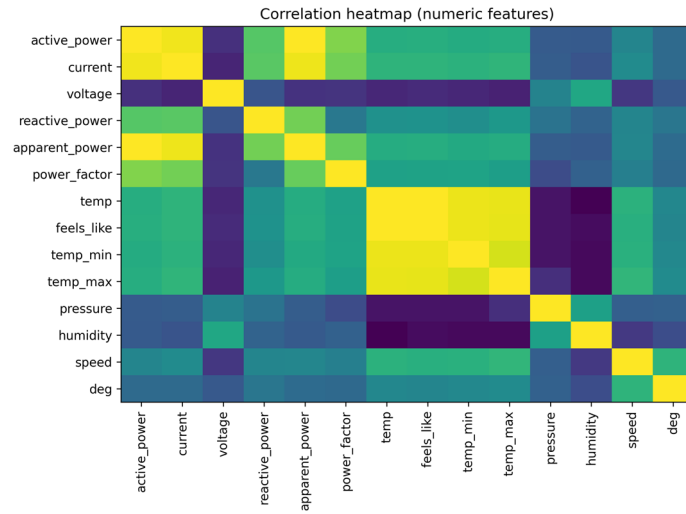


Fig. 6 Correlation heatmap for numeric variables (target and covariates)

temp, feels_like, temp_min, temp_max), which motivates using regularization (dropout) and leakage-free evaluation to ensure robust generalization.

Multi-horizon forecasting formulation: direct multi-output strategy

We adopt a *direct multi-output* (multi-horizon) strategy: given an input window of length w hours, the model predicts the entire horizon vector in one forward pass rather than recursively iterating one-step-ahead predictions. For a forecast origin t , the input is

$$X_t = [x_{t-w+1}, \dots, x_t] \in \mathbb{R}^{w \times d}, \quad (3)$$

and the model outputs

$$\hat{y}_t = [\hat{y}_{t+1}, \dots, \hat{y}_{t+h}] \in \mathbb{R}^h, \quad (4)$$

for $h \in \{24, 168\}$. This formulation reduces error accumulation typical of recursive strategies and is aligned with operational needs where the full next-day/next-week profile is required for planning and scheduling.

Preprocessing and normalization

We performed systematic preprocessing to ensure data integrity and reproducibility. Timestamps were parsed and sorted to obtain a strictly chronological hourly series. We verified that the dataset contains no missing values, no duplicate timestamps, and no missing hourly time steps over the study period.

All continuous variables were scaled using min-max normalization [22]:

$$\tilde{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \quad (5)$$

Normalization parameters ($x_{\min}, x_{\max}$) were fitted *only* on the training split and then applied to validation and test splits to prevent information leakage.

Feature set and target definition

The forecasting target is hourly active power consumption (`active_power`). The input feature set contains $d = 13$ numerical covariates: electrical measurements (`current`, `voltage`, `reactive_power`, `apparent_power`, `power_factor`) and weather variables (`temp`, `feels_like`, `temp_min`, `temp_max`, `pressure`, `humidity`, `speed`, `deg`).

All variables were screened for leakage. In particular, any forward-shifted fields (e.g., `temp_t+1`) were excluded from the training inputs to ensure that only information available up to the forecast origin is used.

Windowing and data splits

We cast forecasting as a supervised multi-output sequence-to-sequence problem. Using a sliding input window of length L , the model predicts the next H time steps:

$$\underbrace{X_{t-L+1:t}}_{\text{input window}} \rightarrow \underbrace{Y_{t+1:t+H}}_{\text{multi-step target}}, \quad (6)$$

where $H \in \{24, 168\}$ (24 h and 168 h ahead).

Data were split chronologically into training/validation/testing subsets (70%/15%/15%). The validation split was used exclusively for hyperparameter tuning and model selection, while the test split remained fully held out for final evaluation.

Proposed hybrid transformer-BiLSTM model

We propose a hybrid Transformer-BiLSTM forecaster for multi-output household load prediction at two horizons, $H \in \{24, 168\}$ hours. The model combines (i) a Transformer encoder to capture long-range dependencies and multi-scale temporal interactions within the historical window and (ii) a BiLSTM refinement block to model local sequential dynamics and smooth transitions. The output is produced in a *direct multi-output* manner, i.e., the full vector $(t+1, \dots, t+H)$ is predicted in one forward pass, avoiding recursive error accumulation.

Notation and input-output mapping

Let L denote the input window length (hours) and d the number of covariates. For each forecast origin t , the model receives $X_t = [x_{t-L+1}, \dots, x_t] \in \mathbb{R}^{L \times d}$ and predicts $\hat{Y}_t = [\hat{y}_{t+1}, \dots, \hat{y}_{t+H}] \in \mathbb{R}^H$. Because windows end at time t and splits are chronological (Section "Windowing and data splits"), the architecture remains leakage-free even though the BiLSTM scans the *past window* in both directions.

Transformer encoder for global temporal context

Each input vector $x_\tau \in \mathbb{R}^d$ is mapped to a d_{model} -dimensional embedding and augmented with positional encodings:

$$H_0 = \text{Embed}(X_t) + P \in \mathbb{R}^{L \times d_{\text{model}}}. \quad (7)$$

A Transformer encoder stack then produces contextual representations $H_E \in \mathbb{R}^{L \times d_{\text{model}}}$ using multi-head self-attention and a position-wise feedforward network:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}, \quad (8)$$

$$\mathbf{H}' = \text{LayerNorm}(\mathbf{H} + \text{Dropout}(\text{MHA}(\mathbf{H}))), \quad (9)$$

$$\mathbf{H}_E = \text{LayerNorm}(\mathbf{H}' + \text{Dropout}(\text{FFN}(\mathbf{H}'))). \quad (10)$$

BiLSTM refinement for local sequential dynamics

To refine the encoded sequence with gated recurrence, we pass $\mathbf{H}_E$ through a BiLSTM:

$$\mathbf{Z} = \text{BiLSTM}(\mathbf{H}_E) \in \mathbb{R}^{L \times 2u}, \quad (11)$$

where u is the number of hidden units per direction. This stage helps model local ramps and short-term persistence patterns that are common in household demand.

Additive (Bahdanau) attention pooling

Instead of compressing the entire window into a single last-state vector, we apply additive attention to form a context vector:

$$\mathbf{e}_\tau = \mathbf{v}^\top \tanh(\mathbf{W}\mathbf{Z}_\tau + \mathbf{b}), \quad (12)$$

$$\alpha_\tau = \frac{\exp(\mathbf{e}_\tau)}{\sum_{j=1}^L \exp(\mathbf{e}_j)}, \quad \mathbf{c} = \sum_{\tau=1}^L \alpha_\tau \mathbf{Z}_\tau \in \mathbb{R}^{2u}. \quad (13)$$

The weights $\{\alpha_\tau\}$ provide a transparent indication of which historical periods contribute most to the forecast.

Multi-output forecasting head (direct strategy)

A compact representation is formed and mapped to horizon-specific outputs:

$$\mathbf{z} = \text{Dropout}(\text{ReLU}(\mathbf{W}_z \mathbf{c} + \mathbf{b}_z)). \quad (14)$$

For each horizon $H \in \{24, 168\}$, we train a dedicated output head:

$$\hat{\mathbf{y}}^{(H)} = \mathbf{W}_H \mathbf{z} + \mathbf{b}_H \in \mathbb{R}^H. \quad (15)$$

Training separate horizon-specific models improves stability and simplifies deployment while preserving a consistent backbone architecture.

Training objective and optimization

Models are trained by minimizing mean squared error on normalized targets:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{H} \sum_{i=1}^H (y_{t+i} - \hat{y}_{t+i})^2. \quad (16)$$

Hyperparameters (e.g., d_{model} , number of encoder layers, number of heads, BiLSTM units u , dropout, and learning rate) are selected using Bayesian optimization on the validation set (Table 2).

Table 2 Optuna-based Bayesian optimization (TPE) procedure used for hyperparameter tuning

Step	Description
1	Initialize an Optuna study and evaluate an initial set of configurations r sampled from $\mathcal{R}$.
2	Fit/update the TPE density models on observed pairs $\{(r, f(r))\}$ and propose a new r (expected improvement).
3	Train the model for r with early stopping; compute $f(r)$ as validation MASE under rolling-origin evaluation.
4	Repeat until the trial budget is exhausted; return the incumbent r^* .
5	Retrain using train+validation data under r^* and report final performance on the held-out test set.

Model dimensions (for reproducibility)

In our setup, $d = 13$ numeric covariates are used. Hence, for each forecast origin, the input tensor has shape (L, d) ; after embedding (L, d_{model}) ; after the encoder (L, d_{model}) ; after BiLSTM $(L, 2u)$; attention pooling yields $c \in \mathbb{R}^{2u}$; and the output head returns $\hat{y}^{(H)} \in \mathbb{R}^H$ for $H \in \{24, 168\}$.

Computational complexity (Big-O)

We report asymptotic computational and memory complexity using explicit model variables rather than a generic “Big- $O(n)$ ” statement. Let L denote the input window length (sequence length), d the input feature dimension, d_{model} the Transformer embedding width, n_h the number of attention heads with head dimension d_k (so $d_{\text{model}} = n_h d_k$), d_{ff} the feed-forward hidden width, u the number of BiLSTM units per direction, B the batch size, and N_{blk} the number of Transformer blocks.

Transformer encoder. For one multi-head self-attention (MHSA) layer, forming attention scores requires $QK^\top$ with cost

$$\mathcal{O}(B n_h L^2 d_k) = \mathcal{O}(B L^2 d_{\text{model}}),$$

while the attention-weighted values contribute the same order. Linear projections and residual/normalization add $\mathcal{O}(B L d_{\text{model}}^2)$, and the position-wise feed-forward network adds $\mathcal{O}(B L d_{\text{model}} d_{\text{ff}})$ per block. Thus, per block the dominant term is typically the quadratic attention term when L is large:

$$\mathcal{O}(B L^2 d_{\text{model}} + B L d_{\text{model}} d_{\text{ff}}).$$

Memory is dominated by storing the attention matrix, i.e.,

$$\mathcal{O}(B n_h L^2)$$

per block (plus activations $\mathcal{O}(B L d_{\text{model}})$).

BiLSTM refinement. A (bi)LSTM layer processes the sequence in $\mathcal{O}(L)$ recurrent steps. With input width p (here $p = d_{\text{model}}$) and hidden size u , the per-layer time complexity is

$$\mathcal{O}(B L (u p + u^2)),$$

where $u p$ accounts for input-to-hidden projections and u^2 for hidden-to-hidden transitions (four gates absorbed into constants). The corresponding activation memory scales as $\mathcal{O}(B L u)$ (per direction).

Overall. For the proposed hybrid, the forward-pass complexity is the sum of the encoder and BiLSTM terms:

$$\mathcal{O}(N_{\text{blk}}(B L^2 d_{\text{model}} + B L d_{\text{model}} d_{\text{ff}}) + B L (u d_{\text{model}} + u^2)),$$

with peak memory dominated by $\mathcal{O}(N_{\text{blk}} B n_h L^2)$ from attention. This explains why throughput decreases as larger windows ($L \in \{168, 336\}$) or wider embeddings are used, consistent with the latency/throughput measurements in Sect. "Inference efficiency (latency and throughput)".

Bayesian optimization for hyperparameter tuning

Deep learning performance in load forecasting is sensitive to hyperparameters that govern optimization (learning rate, batch size), model capacity (depth, hidden units, embedding size), and regularization (dropout). To tune these efficiently and reproducibly under a fixed computational budget, we use *Bayesian optimization* implemented with **Optuna**. Optuna performs sequential model-based optimization using a **Tree-structured Parzen Estimator (TPE)** sampler, which is well-suited to mixed discrete/continuous search spaces common in neural architectures.

Let $r \in \mathcal{R}$ denote a hyperparameter configuration and $f(r)$ the validation objective computed under the leakage-free, chronological protocol (rolling-origin validation). We select the configuration that minimizes the validation loss:

$$r^* = \arg \min_{r \in \mathcal{R}} f(r), \quad (17)$$

where in this study $f(r)$ is the mean validation **MASE** (with MAE/RMSE tracked as secondary metrics).

Optuna TPE search and pruning.

At each iteration, Optuna updates its TPE models of “good” vs. “bad” configurations based on past trials and proposes the next configuration by maximizing expected improvement. Each trial trains the model with early stopping on the validation set. To reduce unnecessary computation, we optionally enable Optuna *pruning* (e.g., median pruning), which terminates trials that are unlikely to outperform the current best based on intermediate validation scores.

Learning-rate policy: The learning rate is treated as a tunable hyperparameter in Optuna. If a fixed learning rate is used, Adam runs with the BO-selected constant value. If a learning-rate scheduler is used (e.g., ReduceLROnPlateau), we report the scheduler type and its parameters (and include the learning-rate curve vs. epoch if requested).

Baseline rationale and fairness controls

We compare against representative model families commonly used in energy forecasting: recurrence (BiLSTM), convolution (TCN), and attention (Transformer), plus N-BEATS as a strong direct multi-output forecaster for short horizons. To ensure fairness, all learning-based models use identical leakage-free chronological splits, training-only normalization, the same rolling origin evaluation protocol, the same early stopping rule, and the same Optuna trial budget.

Forecast evaluation metrics (MAE, RMSE, MAPE)

Consistent and leakage-free evaluation is essential in energy forecasting, where errors translate into operational and planning impacts. Following standard practice [23, 24], we

report three complementary point-forecast metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Let a forecast be issued at origin time t , with horizon $h \in \{24, 168\}$. Denote the ground-truth trajectory by $\{y_{t+i}\}_{i=1}^h$ and predictions by $\{\hat{y}_{t+i}\}_{i=1}^h$. Define the step-ahead errors

$$e_{t+i} = y_{t+i} - \hat{y}_{t+i}, \quad i = 1, \dots, h. \quad (18)$$

For each origin t , the horizon-averaged MAE and RMSE are

$$\text{MAE}_t = \frac{1}{h} \sum_{i=1}^h |e_{t+i}|, \quad \text{RMSE}_t = \sqrt{\frac{1}{h} \sum_{i=1}^h e_{t+i}^2}. \quad (19)$$

Across N rolling forecast origins, the aggregate metrics are

$$\text{MAE} = \frac{1}{N} \sum_{t=1}^N \text{MAE}_t, \quad \text{RMSE} = \sqrt{\frac{1}{Nh} \sum_{t=1}^N \sum_{i=1}^h e_{t+i}^2}. \quad (20)$$

MAPE summarizes relative error in percentage terms. For each origin t , we compute

$$\text{MAPE}_t = \frac{100}{h} \sum_{i=1}^h \frac{|e_{t+i}|}{|y_{t+i}| + \varepsilon}, \quad (21)$$

and aggregate across origins as

$$\text{MAPE} = \frac{1}{N} \sum_{t=1}^N \text{MAPE}_t, \quad (22)$$

where $\varepsilon > 0$ is a small constant used to avoid instability when y_{t+i} is close to zero.

Interpretation. MAE reflects typical absolute deviation and is easy to relate to operational scheduling; RMSE penalizes large deviations and is informative for peak-sensitive planning; MAPE provides a scale-normalized percentage error. For multivariate targets, metrics can be computed per series and horizon and then macro-averaged to avoid domination by high-variance components. Statistical differences between competing models are assessed using Diebold-Mariano tests on horizon-aligned loss differentials (Sect. "[Statistical significance of forecast accuracy \(Diebold–Mariano Test\)](#)"). MASE was used for validation/early stopping because it is robust and scale-free, while MAE/RMSE/MAPE are reported for comparability with prior load-forecasting studies.

Explainability and sensitivity analysis

To support trustworthy deployment in energy informatics, we complement accuracy evaluation with (i) *explainability* of forecast drivers and (ii) *sensitivity/robustness* under operational stress regimes and degraded inputs. Explainability is assessed using SHapley Additive exPlanations (SHAP) and attention diagnostics; sensitivity is assessed by stratifying errors across temperature and ramp periods and by perturbation tests (noise and feature dropout).

SHAP explanations (global and local)

We employ SHAP [16] to attribute model outputs to input features. For a feature vector $\mathbf{x} \in \mathbb{R}^d$ and trained predictor f , SHAP uses an additive attribution model:

$$f(\mathbf{x}) = \mathbb{E}_{\mathbf{X}}[f(\mathbf{X})] + \sum_{j=1}^d \phi_j(\mathbf{x}), \quad (23)$$

where $\phi_j(\mathbf{x})$ is the Shapley value for feature j and $\mathbb{E}[f(\mathbf{X})]$ is the baseline prediction under a background distribution. The Shapley value is defined as the average marginal contribution of feature j over all coalitions:

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq M \setminus \{j\}} \frac{|S|!(d - |S| - 1)!}{d!} \left(f_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - f_S(\mathbf{x}_S) \right), \quad (24)$$

with $M = \{1, \dots, d\}$.

Global importance. For N^* test instances, we summarize feature importance as the mean absolute SHAP value:

$$I_j = \frac{1}{N^*} \sum_{n=1}^{N^*} |\phi_j(\mathbf{x}^{(n)})|. \quad (25)$$

If features are grouped (e.g., weather vs. electrical variables), group importance is

$$I_g = \frac{1}{N^*} \sum_{n=1}^{N^*} \sum_{j \in \mathcal{G}_g} |\phi_j(\mathbf{x}^{(n)})|, \quad (26)$$

where $\mathcal{G}_g$ is the index set for group g .

Local explanations. For a single forecast origin, SHAP provides an additive decomposition:

$$f(\mathbf{x}) - \mathbb{E}[f(\mathbf{X})] = \sum_{j=1}^d \phi_j(\mathbf{x}), \quad (27)$$

visualized with waterfall/force plots to show which variables push the prediction up or down.

Multi-output aggregation. Because the model outputs a horizon vector $\hat{\mathbf{y}} \in \mathbb{R}^h$ (multi-output forecasting), we compute SHAP per horizon step and summarize by averaging absolute contributions across steps:

$$\bar{\phi}_j(\mathbf{x}) = \frac{1}{h} \sum_{i=1}^h |\phi_{j,i}(\mathbf{x})|. \quad (28)$$

This yields a single attribution per feature for each forecast origin, facilitating both local and global interpretation for $h \in \{24, 168\}$.

Attention diagnostics

For the transformer-based components, we extract attention weight matrices from the encoder self-attention heads to assess which historical time steps and variables are

emphasized. We summarize attention by (i) averaging weights across heads and layers and (ii) aggregating across test origins to obtain stable temporal saliency patterns. These diagnostics provide a complementary, model-internal view of temporal focus, while SHAP provides model-agnostic feature attribution.

Sensitivity under operational regimes and degraded inputs

Temperature-regime sensitivity. To assess heat-stress behavior, we partition forecast origins into temperature quantiles (Q1–Q4) using the contemporaneous temperature feature (or daily mean temperature) and report error metrics per quantile. This reveals whether models degrade disproportionately in hot regimes associated with peaks.

Ramp-period sensitivity. To evaluate performance during rapid demand changes, we analyze errors by hour-of-day and by ramp magnitude. Ramp magnitude is defined from the target series as

$$\Delta y_t = y_t - y_{t-1}, \quad (29)$$

and “ramp” periods can be operationally defined as the top q -quantile of $|\Delta y_t|$ (e.g., $q = 0.90$). Errors are then summarized separately for ramp vs. non-ramp periods and by hour-of-day.

Noise robustness. We test robustness to measurement/forecast noise by adding scale-aware Gaussian perturbations to each feature channel:

$$\tilde{x} = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \text{diag}(\lambda^2 \hat{\sigma}^2)), \quad (30)$$

where $\hat{\sigma}$ are per-feature standard deviations computed on the training set (after normalization if applicable) and λ controls noise intensity.

Feature-dropout robustness. To emulate missing sensors or unavailable covariates, we apply channel-wise masking with drop rate p :

$$\tilde{x}_j = \begin{cases} 0, & \text{with probability } p, \\ x_j, & \text{with probability } 1 - p, \end{cases} \quad j = 1, \dots, d, \quad (31)$$

applied after normalization. Performance is reported under clean, noise, and feature-dropout conditions, along with relative degradation versus clean input.

Economic and operational case study

To examine how forecasting accuracy propagates into downstream energy decisions, we embed the proposed forecasting models into a simplified household energy management scenario with battery storage and time-of-use (TOU) tariffs. The purpose of this case study is *methodological*: to illustrate the interaction between forecast characteristics and operational decisions, rather than to derive an optimal control policy.

Household setting and tariff structure. The analysis considers a single residential customer equipped with a behind-the-meter battery. Electricity consumption is measured at hourly resolution, and a representative 168-hour (one-week) segment is selected from the test set. A three-level TOU tariff is assumed, comprising off-peak, mid, and peak periods defined by fixed hourly ranges within the day. Energy costs are computed as the product of hourly grid imports and the corresponding tariff rate.

Battery model. The battery is characterized by a finite energy capacity (kWh), a maximum charge/discharge power (kW), round-trip efficiency, and minimum and maximum

state-of-charge (SoC) limits. Battery dynamics are simulated on an hourly basis, enforcing physical constraints on SoC evolution, charging and discharging power, and efficiency losses. Battery degradation, export remuneration, and network constraints are not considered, consistent with the illustrative scope of the study.

Forecast-driven heuristic controller. Battery operation is governed by a rule-based, forecast-sensitive heuristic controller. At each hour, the controller determines charging or discharging actions based solely on: (i) the model-generated load forecast for the upcoming horizon, (ii) the current tariff level, and (iii) the current battery SoC.

For each forecasting model, low- and high-demand regimes are identified using quantiles of the model's own forecasted load distribution over the planning horizon. These model-specific thresholds define "cheap" and "expensive" demand periods from the controller's perspective. Charging is encouraged during forecasted low-demand or off-peak periods, particularly when peak tariff hours are anticipated ahead, while discharging is prioritized during forecasted high-demand or peak tariff periods. Controller aggressiveness is modulated by normalized forecast deviations and limited by battery power and SoC constraints. Importantly, control actions depend exclusively on forecast information and tariff signals, ensuring a fully forecast-driven operational setup without access to future realized demand.

Baseline and evaluation quantities. As a reference, a no-battery baseline is simulated in which all household demand is met directly from the grid at TOU prices. For each strategy, the simulation records hourly grid imports, battery SoC trajectories, and cumulative energy cost over the representative week. These quantities are later used for comparative analysis in the Results section.

Scope and limitations. The proposed controller is intentionally heuristic and does not solve an optimization or model predictive control (MPC) problem. Its role is to provide a transparent and reproducible mechanism for assessing how differences in forecast behavior influence operational outcomes. Consequently, results from this case study should be interpreted as illustrative rather than economically optimal, and more advanced control formulations are left for future work.

Workflow pseudocode and flowchart

To improve reproducibility and clarify the end-to-end pipeline, Algorithm 1 summarizes the full methodology from data preparation to evaluation, explainability, robustness, and operational embedding. Figure 7 presents a conceptual map of the study workflow, highlighting the main stages from data preparation through forecasting, validation, explainability, robustness testing, and operational embedding.

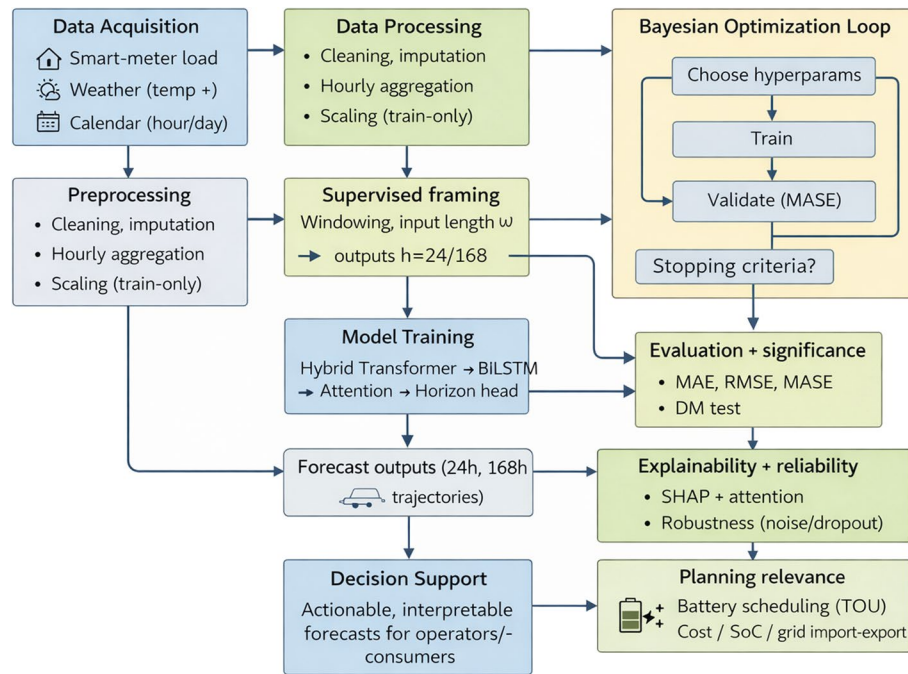


Fig. 7 Conceptual map of the proposed end-to-end forecasting framework: data acquisition and preprocessing, multi-horizon forecasting, evaluation and statistical testing, explainability, robustness/efficiency analysis, and planning-oriented case study

Require: Dataset $\mathcal{D} = \{(\mathbf{x}_t, y_t)\}_{t=1}^T$, $\mathbf{x}_t \in \mathbb{R}^d$, $y_t \in \mathbb{R}$
Require: Horizons $\mathcal{H} = \{24, 168\}$, window length w , model class $\mathcal{M}$, tuning budget B

Ensure: Trained models $\{m_h^*\}_{h \in \mathcal{H}}$, metrics, DM tests, SHAP values, robustness and planning traces

PHASE I: DATA PREPARATION (NO LEAKAGE)

- 1: Parse timestamps; enforce hourly regularity; remove duplicates
 - 2: Define target $y_t \leftarrow$ active power; retain inference-available features
 - 3: Chronological split: $\mathcal{D}_{\text{train}}$ (70%), $\mathcal{D}_{\text{val}}$ (15%), $\mathcal{D}_{\text{test}}$ (15%)
 - 4: Fit scaler/encoders on $\mathcal{D}_{\text{train}}$ only; transform $\mathcal{D}_{\text{val}}$, $\mathcal{D}_{\text{test}}$
-

PHASE II: MODEL TRAINING AND SELECTION (BAYESIAN OPTIMIZATION)

- 5: **for** $h \in \mathcal{H}$ **do**
 - 6: Build windows: $\mathbf{X}^{(n)} = [\mathbf{x}_{t-w+1}, \dots, \mathbf{x}_t]^\top$, $\mathbf{y}^{(n)} = [y_{t+1}, \dots, y_{t+h}]^\top$
 - 7: **for** $b = 1, \dots, B$ **do**
 - 8: Sample hyperparameters θ_b (Optuna)
 - 9: Train $m_b \in \mathcal{M}$ with early stopping on $\mathcal{D}_{\text{val}}$
 - 10: $\ell_b \leftarrow \text{MASE}(m_b, \mathcal{D}_{\text{val}})$ $\triangleright$ objective; report MAE/RMSE/MAPE later
 - 11: Update surrogate; choose next θ_{b+1} via acquisition
 - 12: **end for**
 - 13: $\theta^* \leftarrow \arg \min_b \ell_b$
 - 14: Retrain m_h^* on $\mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}}$ using θ^*
 - 15: **end for**
-

PHASE III: TEST EVALUATION AND ANALYSIS (ROLLING-ORIGIN)

- 16: **for** $h \in \mathcal{H}$ **do**
 - 17: Generate rolling-origin forecasts on $\mathcal{D}_{\text{test}}$: $\hat{\mathbf{Y}} \leftarrow m_h^*(\cdot)$
 - 18: Compute MAE, RMSE, MAPE (and optional per-step curves) aggregated over origins
 - 19: Perform Diebold–Mariano tests vs. baselines using aligned loss series
 - 20: Compute SHAP values; aggregate global feature importance
 - 21: Robustness: evaluate Clean, Noise (λ -scaled), Feature-dropout (p) $\Rightarrow \Delta \text{RMSE}$
 - 22: **end for**
-

PHASE IV: PLANNING CASE STUDY (DECISION-CENTRIC)

- 23: Simulate (i) No Battery, (ii) TOU-rule (forecast-agnostic), (iii) forecast-informed heuristic (per model)
 - 24: Record SoC trajectories, grid exchanges, and total cost; aggregate across rolling 168 h windows
 - 25: **return** $\{m_h^*\}_{h \in \mathcal{H}}$, metrics, DM tests, SHAP summaries, robustness and planning logs
-

Algorithm 1 Leakage-free multi-horizon forecasting pipeline with evaluation, explainability, robustness, and planning

Results

We report predictive performance for *short-term* ($h=24$ h) and *medium-term* ($h=168$ h) horizons under a leakage-free, time-ordered evaluation protocol. All preprocessing operations that learn from data (feature scaling, target scaling, and any encoding) are fitted on the training portion of each split only and then applied unchanged to validation and test windows. Models are trained in a multi-output setting, producing the full trajectory $\hat{\mathbf{y}} \in \mathbb{R}^h$ in a single forward pass. Hyperparameters are selected via Bayesian optimization with matched search budgets across models, and training is regularized with early stopping based on validation MASE.

Table 3 Optuna search space and best setting (24 h)

Hyperparameter	Search Range/Prior	Optimal Value (24h)
Attention heads h	[2, 8] (integer)	6
Head dimension d_k	{16, 32, 48, 64}	32
FFN hidden size d_{ff}	{64, 128, 192, 256}	128
LSTM units per direction	{32, 64, 96, 128}	128
Dropout (attn/LSTM/proj)	[0.10, 0.50] (uniform)	0.30
Learning rate (Adam)	$[10^{-4}, 10^{-2}]$ (log-uniform)	8.63×10^{-4}
Transformer blocks L	[1, 3] (integer)	1

Note. Discrete parameters are sampled uniformly; the learning rate is sampled log-uniformly. The best setting minimizes the validation objective on the chronological validation split.

Table 4 Optuna search space and best setting (168 h)

Hyperparameter	Search Range/Prior	Optimal Value (168h)
Attention heads h	[2, 8] (integer)	5
Head dimension d_k	{16, 32, 48, 64, 80, 96}	80
FFN hidden size d_{ff}	{128, 192, 256, 320, 384}	256
LSTM units per direction	{64, 96, 128, 160, 192}	96
Dropout (attn/LSTM/proj)	[0.10, 0.50] (uniform)	0.40
Learning rate (Adam)	$[10^{-4}, 5 \times 10^{-3}]$ (log-uniform)	1.15×10^{-3}
Transformer blocks L	[1, 3] (integer)	1

Note. Discrete parameters are sampled uniformly; the learning rate is sampled log-uniformly. The best setting minimizes the validation objective on the chronological validation split.

Results are organized as follows. Section "[Hyperparameter tuning \(Optuna\)](#)" summarizes the tuning outcomes and hyperparameter importance. Section "[Forecasting performance](#)" compares forecasting accuracy across horizons using MAE, RMSE, and MAPE. Section "[Statistical significance of forecast accuracy \(Diebold–Mariano Test\)](#)" reports Diebold–Mariano tests to assess whether observed accuracy differences are statistically significant. Section "[Error structure](#)" analyzes error structure across temperature regimes, forecast steps, and hour-of-day ramps. Section "[Explainability via SHAP](#)" presents explainability findings using SHAP. Section "[Robustness](#)" evaluates robustness to noise and feature dropout, and Sect. "[Planning relevance: decision-centric \(Economic\) evaluation](#)" illustrates planning relevance through a tariff-based battery scheduling case study.

Hyperparameter tuning (Optuna)

Model performance is highly contingent upon architectural and optimization hyperparameters that govern model capacity, regularization, and convergence. To circumvent ad hoc manual tuning, we employ Bayesian hyperparameter optimization via the Optuna framework. For each forecasting horizon ($h \in \{24, 168\}$), we conduct an independent study, evaluating each trial on the validation split using a consistent leakage-free chronological protocol and early stopping (monitored via the validation objective).

The search space encompasses the number of attention heads, head dimension, feed-forward size, BiLSTM hidden units, dropout rate, learning rate (log-scaled), and the number of Transformer blocks. Tables 3 and 4 present the explored ranges and the optimal configurations identified. Figure 8 depicts parameter-importance estimates derived from the Optuna study history, indicating which hyperparameters most significantly influenced validation performance within the explored ranges.

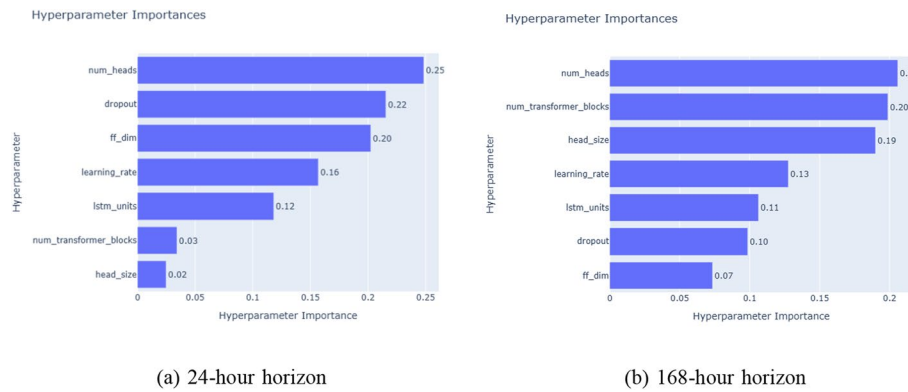


Fig. 8 Optuna-based hyperparameter importance for (a) the 24-hour and (b) the 168-hour forecasting horizons. The rankings summarize how strongly each hyperparameter is associated with validation MASE improvements within the explored search space (importance is model- and search-space dependent)

Table 5 Forecast accuracy for the 24-hour horizon

Model	MAE	RMSE	MAPE (%)
N-BEATS	0.053081	0.068364	29.8628
Hybrid (T-BiLSTM)	0.053943	0.070128	29.7353
BiLSTM	0.063498	0.082086	35.5721
Transformer	0.064442	0.082036	36.6421

Legend. MAE and RMSE are reported in the normalized target scale; MAPE is in percent. Lower values indicate better performance

Table 6 Forecast accuracy for the 168-hour horizon

Model	MAE	RMSE	MAPE (%)
Hybrid (T-BiLSTM)	0.056585	0.074605	30.1067
BiLSTM	0.061459	0.079263	34.0090
TCN	0.067984	0.087580	36.7417
Transformer	0.080273	0.109810	33.9234

Legend. MAE and RMSE are reported in the normalized target scale; MAPE is in percent. Lower values indicate better performance.

Learning-rate setting. The learning rate is treated as a tunable hyperparameter; during any given trial, it is held constant throughout training (using the Adam optimizer). Early stopping is employed to mitigate overfitting.

Interpretation. Across both horizons, attention configuration (e.g., number of heads and head dimension), regularization strength (dropout), and feed-forward width emerge as the most influential parameters. This underscores the critical trade-off between model expressivity and generalization. Importantly, the reported importance scores reflect sensitivity strictly within the specified search space and should not be construed as universal rankings beyond this specific experimental context.

Forecasting performance

We evaluate the proposed Hybrid Transformer-BiLSTM against established deep learning baselines across two forecast horizons: short-term (24 h) and medium-term (168 h). Performance is summarized using MAE, RMSE, and MAPE, which respectively quantify average absolute deviation, sensitivity to large errors (e.g., peaks), and relative percentage error. We report results for each horizon separately (Tables 5 and 6) to account

for the substantial differences in forecasting difficulty and error characteristics between daily and weekly horizons.

Short-term (24-hour) forecasting. Table 5 indicates that N-BEATS achieves the lowest MAE and RMSE, while the Hybrid Transformer-BiLSTM attains the best MAPE and remains competitive with N-BEATS on the former metrics. This suggests that while the hybrid model is a strong performer for intraday prediction, it is not uniformly superior across all error criteria. Relative to the BiLSTM and vanilla Transformer baselines, the hybrid model significantly reduces MAE and RMSE, demonstrating that combining attention-based long-context encoding with recurrent refinement improves the tracking of rapid household fluctuations in the short term. We further verify the statistical significance of these differences in Sect. "Statistical significance of forecast accuracy (Diebold–Mariano Test)" using Diebold–Mariano tests.

Medium-term (168-hour) forecasting. For the weekly horizon (Table 6), the Hybrid Transformer-BiLSTM achieves the lowest MAE, RMSE, and MAPE among all evaluated baselines. The advantage is most pronounced relative to the vanilla Transformer, consistent with the challenge of maintaining stability over long horizons amid regime shifts and accumulated uncertainty. Compared to the BiLSTM and TCN, the hybrid model yields smaller average and peak-sensitive errors, indicating improved generalization over long horizons where demand-weather dependencies unfold over multiple days. To complement these horizon-averaged results, we also report per-step error growth across the 168-hour trajectory (Fig. 11b), illustrating how errors accumulate stepwise rather than solely in aggregate.

Forecast diagnostics. Figures 9 and 10 visualize representative predictions for each horizon. For 24 h, models generally track intraday structure, with remaining discrepancies concentrated around sharper ramps. For 168 h, the hybrid model better preserves

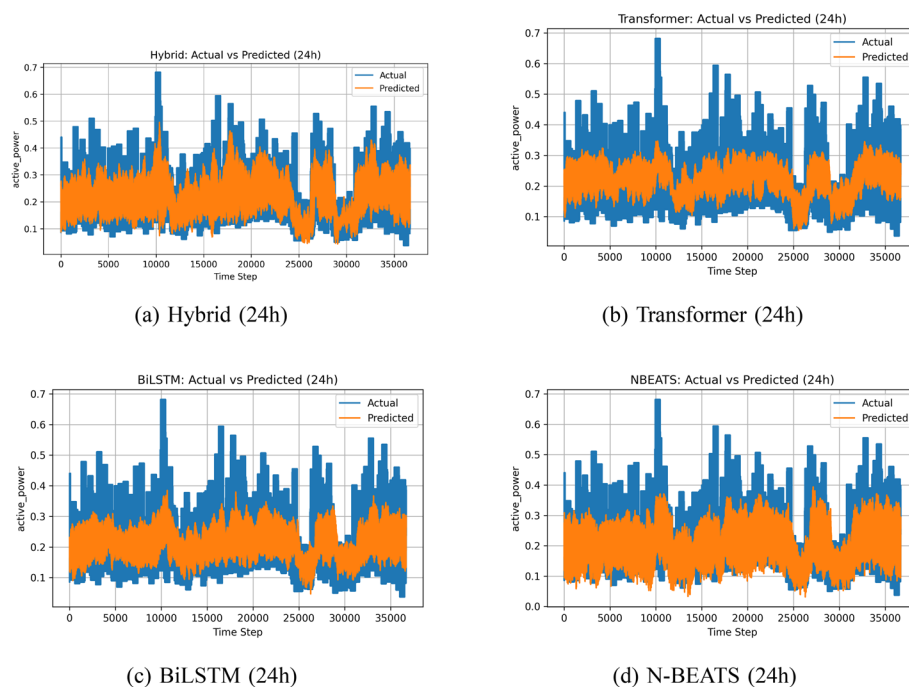


Fig. 9 Forecast diagnostics for the 24-hour horizon. Panels show actual versus predicted trajectories and the corresponding residual series across model variants for a representative test segment

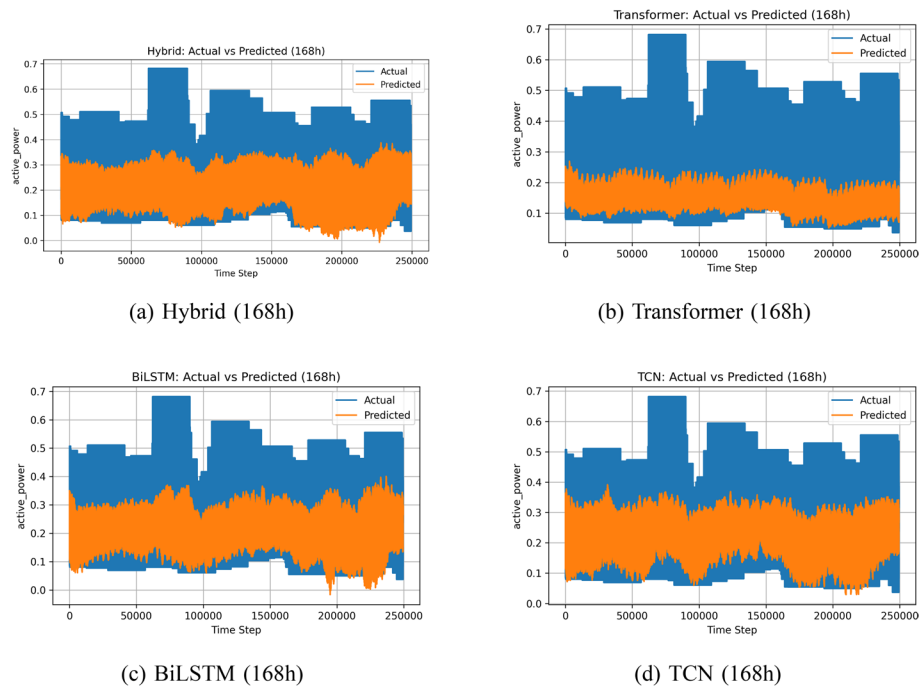


Fig. 10 Forecast diagnostics for the 168-hour horizon. Panels show actual versus predicted trajectories and the corresponding residual series across model variants for a representative test week

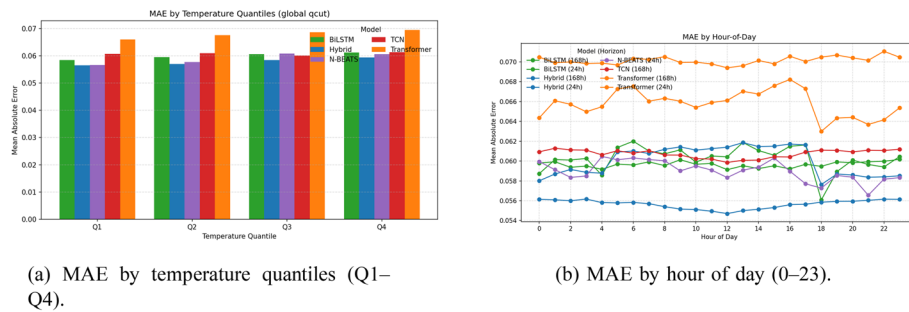


Fig. 11 Error structure diagnostics on the evaluation set. Panel (a) reports MAE stratified by temperature quantiles (Q1–Q4), and panel (b) reports MAE by hour of day (0–23), highlighting error concentration during intraday transitions

the multi-day trend and amplitude, while alternative baselines show larger drift and under/overshoot episodes.

Error structure

Beyond aggregate accuracy, we examine how forecast errors vary across operating regimes that pose challenges to household load forecasting, specifically temperature conditions, forecast lead times, and time-of-day effects. Such diagnostics help determine whether performance gains are uniform or concentrated in specific operational regimes (e.g., ramp events and extreme weather periods).

Temperature regimes

Figure 11 presents MAE stratified by temperature quantiles (Q1–Q4), calculated over the evaluation period using the hourly temperature covariate. Across all models, errors

tend to be higher in warmer regimes (upper quantiles), indicating that demand-weather interactions introduce additional variability that is difficult to model. The Hybrid (Transformer-BiLSTM) exhibits the lowest or near-lowest MAE across all quantiles, with a comparatively smaller degradation in the hottest bin; conversely, the vanilla Transformer demonstrates the highest sensitivity to temperature regime shifts. *Implication.* Regime-stratified performance suggests that evaluation based solely on overall averages can obscure meaningful differences under high-impact conditions (e.g., heat-related load stress). These results underscore the importance of reporting regime-aware diagnostics alongside aggregate metrics.

Hour-of-day effects

Figure 11 reports MAE by hour-of-day. Across models, errors are typically higher during periods associated with rapid consumption shifts (morning and evening transitions) and lower during the more stable nighttime hours. The Hybrid and other sequence-based baselines show tighter clustering throughout the day, whereas the Transformer maintains a consistently higher MAE across most hours. *Implication.* The concentration of errors during intraday transitions highlights the critical need to capture ramp dynamics and time-of-day interactions when forecasting household demand.

Error growth by forecast step

Figure 12 illustrates MAE as a function of lead time over the 168-hour horizon. The Hybrid model exhibits the flattest error trajectory, indicating reduced error accumulation as lead time increases. In contrast, the Transformer baseline displays more pronounced error growth and oscillations over the horizon, consistent with reduced stability at longer lead times in this setting. *Implication.* Minimizing error drift over multi-day horizons is of practical importance for week-ahead scheduling and planning, where cumulative deviations can influence reserve allocation and demand-side decisions.

Overall, the Hybrid Transformer-BiLSTM demonstrates superior stability across temperature regimes, lead times, and intraday cycles. These diagnostics complement the aggregate results and aid in characterizing the specific contexts in which the proposed model offers the greatest operational benefit.

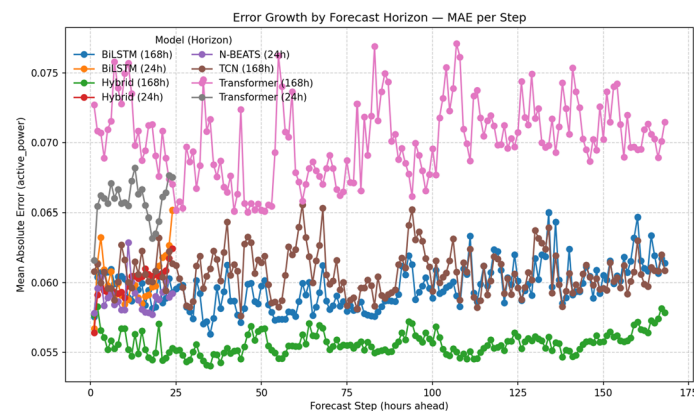


Fig. 12 MAE by lead time across the 168-hour horizon (step-ahead error profile). The Hybrid Transformer-BiLSTM shows the smallest error growth with increasing lead time, indicating improved stability for week-ahead forecasting

Statistical significance of forecast accuracy (Diebold–Mariano Test)

To determine whether differences in forecast accuracy are statistically significant rather than attributable to sampling variability, we employ the Diebold–Mariano (DM) test. This test compares the expected predictive loss of two competing models across aligned forecast origins. Let $\ell(\cdot)$ denote a per-step loss function (absolute error for MAE-based and squared error for MSE-based metrics). For each forecast origin t , we aggregate step-ahead losses within the horizon h to obtain a single loss value per origin:

$$L_t^{(m)} = \frac{1}{h} \sum_{i=1}^h \ell(y_{t+i} - \hat{y}_{t+i}^{(m)}), \quad m \in \{\text{Hybrid, Baseline}\}. \quad (32)$$

We then form the loss differential

$$d_t = L_t^{(\text{Hybrid})} - L_t^{(\text{Baseline})}, \quad (33)$$

so that $d_t < 0$ indicates the Hybrid achieves a smaller average loss (better accuracy) at origin t . The DM statistic tests $H_0 : \mathbb{E}[d_t] = 0$ against a two-sided alternative using a heteroskedasticity and autocorrelation consistent (HAC) variance estimator (Newey–West). Following common practice for multi-step horizons, we set the HAC lag to $L = h - 1$ to account for overlap-induced serial dependence. Table 7 reports DM statistics and p -values comparing the Hybrid Transformer-BiLSTM against each baseline model.

The DM tests were computed using these per-origin aggregated losses, with the HAC lag parameter set to $h - 1$ to account for overlapping forecast errors.

At the 24-hour horizon, the Hybrid model significantly outperforms the Transformer baseline under both MAE-based and MSE-based loss functions ($p < 10^{-6}$). Against the BiLSTM baseline, the Hybrid model demonstrates significant improvement under MAE-based loss ($p = 3.9 \times 10^{-6}$), whereas the MSE-based comparison is marginal ($p = 0.065$). This suggests that the Hybrid model primarily reduces typical errors rather than extreme outliers in the short term. Comparisons with N-BEATS at 24h do not reach statistical significance at the 5% level for either loss function, indicating broadly

Table 7 Diebold–Mariano (DM) tests comparing the Hybrid Transformer-BiLSTM against baseline models. Negative DM values indicate the Hybrid achieves lower average loss (better accuracy)

Horizon	Comparison	Loss	Agg.	$\hat{T}$	DM stat	p -value
24 h	Hybrid vs. Transformer	MAE-like	mean	1529	−23.505	$< 10^{-6***}$
24 h	Hybrid vs. Transformer	MSE-like	mean	1529	−21.784	$< 10^{-6***}$
24 h	Hybrid vs. BiLSTM	MAE-like	mean	1529	−4.621	$3.9 \times 10^{-6***}$
24 h	Hybrid vs. BiLSTM	MSE-like	mean	1529	1.846	0.065
24 h	Hybrid vs. N- BEATS	MAE-like	mean	1529	1.797	0.073
24 h	Hybrid vs. N- BEATS	MSE-like	mean	1529	−0.415	0.679
168 h	Hybrid vs. Transformer	MAE-like	mean	1485	−72.544	$< 10^{-6***}$
168 h	Hybrid vs. Transformer	MSE-like	mean	1485	−70.720	$< 10^{-6***}$
168 h	Hybrid vs. BiLSTM	MAE-like	mean	1485	−68.140	$< 10^{-6***}$
168 h	Hybrid vs. BiLSTM	MSE-like	mean	1485	−64.792	$< 10^{-6***}$
168 h	Hybrid vs. TCN	MAE-like	mean	1485	−54.176	$< 10^{-6***}$
168 h	Hybrid vs. TCN	MSE-like	mean	1485	−48.099	$< 10^{-6***}$

Notes. MAE-like uses $\ell(e) = |e|$ and MSE-like uses $\ell(e) = e^2$. “Agg.” indicates losses are averaged across the h steps within each forecast origin before DM computation. $\hat{T}$ denotes the number of aligned forecast origins. Two-sided DM tests use Newey–West HAC standard errors with lag $L = h - 1$. Negative DM statistics favor the hybrid under the definition $d_t = L_t^{(\text{Hybrid})} - L_t^{(\text{Baseline})}$. Significance codes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

comparable short-term accuracy between the two models under the current evaluation protocol.

Conversely, at the 168-hour horizon, the Hybrid model achieves substantial and highly significant improvements over the Transformer, BiLSTM, and TCN baselines under both loss functions ($p < 10^{-6}$ across comparisons). This indicates that the Hybrid model's accuracy gains at weekly horizons are robust to the choice of loss function (absolute vs. squared error) and persist after adjusting for overlap-induced dependence via HAC standard errors. Operationally, statistically significant reductions in multi-day forecast loss provide more reliable inputs for week-ahead planning (e.g., scheduling and flexibility allocation), particularly when decision-making relies on sustained accuracy over extended lead times.

Inference efficiency (latency and throughput)

In energy informatics, inference efficiency is critical for real-time deployment, particularly within adaptive load forecasting, demand response, and grid management. Latency, denoting the per-sample prediction time (batch size = 1), serves as a primary indicator of real-time responsiveness, whereas throughput measures scalability under batched inference (predictions per second, batch size $B \gg 1$).

Table 8 presents the inference efficiency under two distinct operational regimes. For interactive or real-time settings (batch size = 1), the proposed Hybrid Transformer-BiLSTM achieves low per-sample latency (66–71 ms across horizons) comparable to N-BEATS for the 24 h horizon and substantially lower than the TCN baseline for the 168 h horizon. This suggests that the hybrid model can effectively support near-real-time decision loops (e.g., rolling hourly updates) when deployed as a single-stream predictor.

In large-batch evaluations, throughput exhibits significant divergence across architectures. The hybrid model yields lower throughput compared to convolutional or purely feed-forward baselines. This is expected, as the integration of self-attention and recurrent components introduces operations with a lower degree of parallelism and higher memory traffic per step. Conversely, TCN and N-BEATS leverage highly parallelized kernels during batching, resulting in superior sample rates. Overall, the results indicate that the proposed hybrid framework remains practically deployable (maintaining sub-0.1 s latency), trading batch-scale efficiency for improved accuracy and interpretability in the medium-term horizon.

Explainability via SHAP

While forecasting accuracy is essential, operational deployment in energy systems also necessitates transparent and auditable explanations. Consequently, we employ SHapley Additive exPlanations (SHAP), a cooperative game-theoretic method that attributes

Table 8 Inference efficiency by model and forecast horizon

Model	Horizon	Latency (ms/sample)	Throughput (samples/s)
Hybrid (Transformer-BiLSTM)	24 h	66.3	4604.7
N-BEATS	24 h	68.4	9409.9
Hybrid (Transformer-BiLSTM)	168 h	71.3	1403.6
TCN	168 h	131.1	5691.8

Notes. Latency is measured for batch size 1 and reported as median wall-clock time per sample over repeated runs. Throughput is measured in large-batch mode (using the same hardware/software for all models) and reported as samples per second. Lower latency indicates faster responsiveness; higher throughput indicates better batch processing efficiency.

individual predictions to input features in an additive and locally consistent manner. SHAP complements standard correlation-based analysis by capturing the nonlinear effects and interactions learned by the Hybrid Transformer–BiLSTM model, thereby fostering operator trust and enabling actionable interpretation.

In the 168-hour setting, SHAP values are computed on the held-out test set using the feature representation described in Sect. "Feature set and target definition" (comprising electrical and weather variables, engineered calendar features, and lagged load terms). For multi-step outputs, attributions are calculated for each time step and then aggregated across the horizon to derive a time-averaged importance ranking. Figure 13 illustrates (a) global feature importance and (b) the distribution of signed SHAP impacts across the test instances.

The SHAP analysis (Fig. 13) identifies *voltage* as the dominant driver of medium-term forecasts (21.1%), followed by seasonal proxies (*month*, 15.2%) and meteorological variables (*pressure*, *temp_min*, *temp*). Electrical state variables such as *power_factor*, *apparent_power*, and *current* also contribute materially, indicating that both grid conditions and weather variability shape multi-day demand evolution.

Calendar encodings (*hour*, *dayofweek*) exhibit near-zero marginal contributions in this setup, which is consistent with these effects being captured implicitly by the sequential model dynamics and positional encodings rather than through static categorical features. Overall, the explanation profile supports operational interpretability: the model's multi-day forecasts are primarily governed by a blend of electrical-quality indicators and weather-linked drivers, offering actionable guidance for monitoring and planning.

Robustness

Robustness is critical for deployment, where sensor noise, packet loss, and missing covariates are common. We evaluate robustness via controlled perturbations of the normalized input windows $X_t \in \mathbb{R}^{w \times d}$ under three conditions:

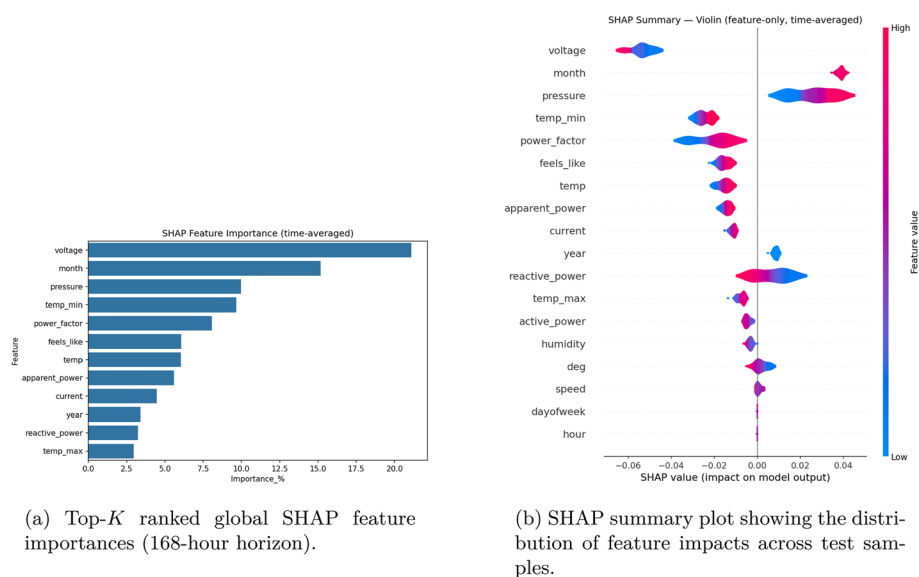


Fig. 13 SHAP-based explainability for the Hybrid Transformer–BiLSTM at the 168-hour horizon. Panel (a) shows time-averaged global feature importance aggregated over the multi-step output, and panel (b) summarizes the signed contribution distributions across test samples (color indicates the feature value)

1. *Clean*: unperturbed inputs.
2. *Noise*: additive, scale-aware Gaussian perturbations applied independently per feature channel,

$$\varepsilon \sim \mathcal{N}(0, \text{diag}(\lambda^2 \hat{\sigma}^2)), \quad \tilde{X}_t = X_t + \varepsilon,$$

where $\hat{\sigma}$ is the per-feature standard deviation estimated on the training split and $\lambda > 0$ controls noise intensity.

3. *Feature dropout*: channel-wise masking applied after normalization,

$$\tilde{X}_{t,:j} = \begin{cases} 0, & \text{with probability } p, \\ X_{t,:j}, & \text{otherwise,} \end{cases} \quad j = 1, \dots, d,$$

where $p \in [0, 1)$ is the channel-drop rate.

Table 9 reports RMSE for each condition and relative degradation $\Delta\text{RMSE} = 100 \times (\text{RMSE}_{\text{cond}} - \text{RMSE}_{\text{clean}}) / \text{RMSE}_{\text{clean}}$.

Noise robustness. RMSE changes are negligible at the tested noise level ($\Delta\text{RMSE} \leq 0.01\%$), indicating that small perturbations are largely attenuated.

Feature dropout. Channel dropout increases error for both models; however, the Hybrid degrades less (+53.82%) than the baseline (+97.67%), suggesting stronger redundancy across covariates.

Planning relevance: decision-centric (Economic) evaluation

Forecasting accuracy is operationally meaningful only insofar as it improves downstream decisions. To quantify economic value without overstating the control contribution, we treat the battery experiment as a *decision-centric evaluation* of forecast utility, not as a proposal for a novel scheduling algorithm.

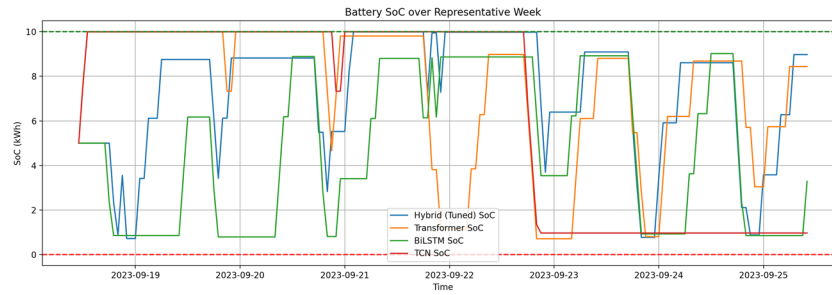
Setup and controller We simulate a residential behind-the-meter battery operating under a fixed time-of-use (TOU) tariff. Hourly demand forecasts drive a transparent heuristic controller subject to state-of-charge (SoC) and power constraints. To isolate the *marginal value of improved forecasts*, the controller logic and all tariff/battery parameters are held constant across models; only the forecast stream changes. The grid-only **No Battery** case is reported as the economic baseline.

Evaluation protocol Planning impact is evaluated over the test period using rolling 168-hour windows. For each window, we compute the total energy cost and savings relative to the **No Battery** baseline. We then aggregate costs and savings across all

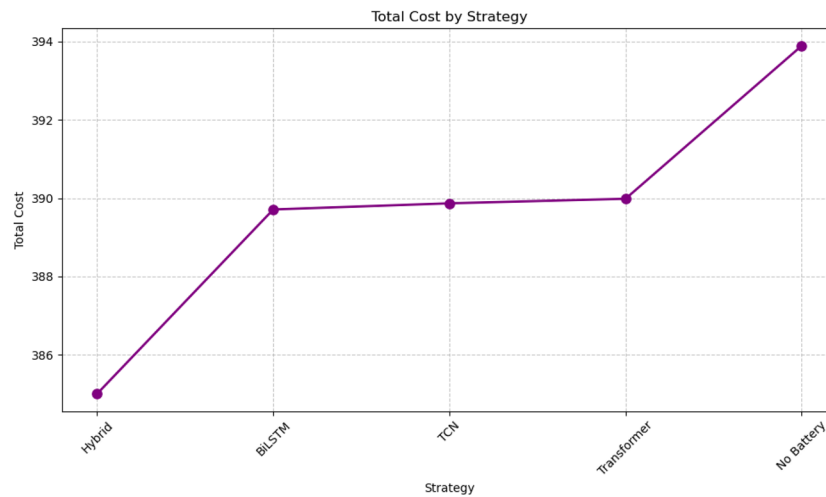
Table 9 Robustness under clean input, scale-aware Gaussian noise, and feature dropout. ΔRMSE is relative to each model's clean-input RMSE. Results shown for $\lambda = 0.05$ and $p = 0.30$.

Model	Condition	RMSE	ΔRMSE (%)
Hybrid (Transformer-BiLSTM)	Clean	0.073	0.00
	Noise	0.073	0.00
	Dropout	0.112	53.82
Transformer (baseline)	Clean	0.087	0.00
	Noise	0.087	0.01
	Dropout	0.172	97.67

Note. Noise uses feature-wise $\hat{\sigma}$ estimated on the training split; feature dropout masks entire channels after normalization. RMSE is computed on the same evaluation windows used for the 168-hour test results



(a) Battery SoC trajectories (representative week).



(b) Total weekly cost by strategy (representative week).

Fig. 14 Decision-centric evaluation of forecast value in household energy management. Panel (a) shows SoC trajectories produced by the *same* heuristic controller when driven by different model forecasts; panel (b) reports the resulting total weekly cost under identical TOU pricing and battery constraints.

Table 10 Energy cost comparison for a representative week under forecast-informed battery scheduling

Strategy	Total Cost	Saving vs. No Battery	% Saving
Hybrid	385.000	8.894	2.258
BiLSTM	389.711	4.184	1.062
TCN	389.868	4.027	1.022
Transformer	389.984	3.910	0.993
No Battery	393.894	0.000	0.000

Note. Results are scenario-conditional (tariff and battery parameters fixed). Export revenue, demand charges, and battery degradation are not modeled; values should be interpreted comparatively within this study

evaluation windows to obtain a scenario-conditional estimate of economic benefit under fixed tariff and battery assumptions.

Representative-week illustration Figure 14 shows typical SoC trajectories and the corresponding total weekly cost for a representative window, while Table 10 reports the numerical cost breakdown. In this example, the hybrid-driven strategy yields the lowest cost among forecast-informed variants.

Aggregated multi-week results Table 11 summarizes costs aggregated across all rolling 168-hour windows in the test set. All forecast-informed strategies reduce cost

Table 11 Energy cost comparison aggregated over the test period using rolling 168-hour windows under a fixed TOU tariff and battery specification

Strategy	Total Cost	Saving vs. No Battery	% Saving
Hybrid	1539.450	36.130	2.293
TCN	1543.759	31.819	2.020
BiLSTM	1559.265	16.312	1.035
Transformer	1569.580	5.998	0.381
No Battery	1575.577	0.000	0.000

Note. Costs are aggregated across all evaluation windows in the test period. Savings are computed relative to the **No Battery** baseline under identical tariff and battery constraints. Export revenue, demand charges, and battery degradation are not modeled; values should be interpreted comparatively within this scenario

relative to the **No Battery** baseline, confirming the economic value of TOU arbitrage when storage is available. Among model-driven variants, the Hybrid forecast yields the largest cumulative savings (2.29%), reducing total cost from 1575.58 to 1539.45. TCN is a close second (2.02%), whereas the Transformer provides only marginal benefit (0.38%), indicating that forecast bias/variance translates directly into operational cost under the fixed controller. While absolute savings depend on tariff structure and battery sizing, the ranking is consistent with the forecasting results: lower long-horizon error tends to produce better charge–discharge timing and higher realized savings.

Limitations This experiment is intentionally lightweight and does not claim optimal control. It quantifies the *incremental economic value* of improved forecasts under a fixed, transparent heuristic, which is appropriate for planning relevance in energy informatics. Future work can replace the heuristic with MPC or reinforcement learning and incorporate export pricing, demand charges, and degradation to enable a fuller techno-economic assessment.

Discussion

The empirical results indicate that the proposed Hybrid Transformer-BiLSTM, optimized via Bayesian optimization, provided an efficient, reproducible way to identify strong configurations under a fixed budget in medium-term forecasting while maintaining competitiveness at short horizons. This section synthesizes findings regarding predictive performance, statistical significance, explainability, robustness, and operational relevance. Furthermore, it outlines the study's limitations and proposes future research directions for energy informatics applications.

Model design and optimization insights

Bayesian optimization (via Optuna) demonstrated efficacy in tuning a high-dimensional hyperparameter space within a fixed computational budget. The joint exploration of attention configuration (number of heads, head size, feed-forward width), recurrent capacity (BiLSTM units), and training dynamics (learning rate, dropout, batch size) yielded well-calibrated models without the need for manual intervention. Analysis of the resulting hyperparameter importance suggests that attention structure and regularization are particularly influential, whereas excessive depth offers limited marginal benefits for household-level series. These results support the perspective that meticulous architecture calibration can outperform naive scaling of model size, particularly given the noisy and non-stationary nature of household-level data.

Forecasting accuracy and statistical significance

At the 24-hour horizon, the Hybrid model performs competitively against strong baselines, notably N-BEATS, with performance variations depending on the selected metric (e.g., MAE/RMSE versus MAPE). Conversely, at the 168-hour horizon, the Hybrid model achieves the highest overall accuracy among the evaluated deep learning baselines. This aligns with the model's design objective: combining long-range dependency modeling via self-attention with local temporal refinement via recurrence.

Diebold-Mariano tests corroborate that the performance improvements of the Hybrid model are statistically robust against Transformer and TCN baselines ($p < 0.001$), showing strong significance at the weekly horizon. At the 24 h horizon, test results indicate that certain differences (e.g., under squared-error loss compared to BiLSTM/N-BEATS) are smaller and potentially statistically indistinguishable. This reinforces the notion that short-horizon performance is often dominated by intraday patterns which multiple architectures can capture effectively.

Model interpretability and error dynamics

SHAP analysis attributes the variance in medium-term forecasts primarily to voltage and weather-related variables (e.g., pressure and minimum temperature), which jointly account for a substantial portion of total feature importance. This alignment with physical expectations confirms that electrical operating conditions and meteorological drivers jointly shape residential demand, particularly over multi-day horizons. Conversely, calendar encodings (e.g., hour, day-of-week) contribute minimally, suggesting that the sequential encoder internalizes temporal regularities without relying heavily on static categorical proxies.

Error-structure diagnostics reveal that forecast degradation is concentrated during (i) high-temperature regimes and (ii) rapid ramp periods—conditions that coincide with operationally stressful grid states. Notably, the Hybrid model exhibits flatter error growth over the 168-hour trajectory and reduced sensitivity to extreme temperature quantiles relative to the vanilla Transformer. This indicates improved long-horizon stability, a critical property for medium-term scheduling and reserve planning.

Robustness and reliability under perturbations

Robustness experiments demonstrate that mild, scale-aware perturbations result in negligible RMSE changes, whereas feature dropout induces significant degradation across all models. Crucially, the Hybrid model degrades substantially less than the comparator models under dropout, suggesting superior cross-channel redundancy and implicit regularization arising from the attention-recurrence combination. This distinction is operationally significant, as household forecasting deployments frequently encounter missing sensors, delayed weather updates, or communication interruptions. Consequently, robustness complements accuracy by enhancing forecast reliability under imperfect input conditions.

Economic effect analytics and planning relevance

To quantify practical value beyond standard prediction metrics, forecasts were integrated into a forecast-informed battery scheduling simulation operating under a fixed time-of-use tariff and identical battery constraints. The economic evaluation was

conducted over multiple aggregated test weeks, reporting total energy cost, absolute savings, and percentage savings relative to the no-battery baseline. Across all models, forecast-informed control reduced costs compared to the baseline, confirming that even approximate forecasts can enable effective tariff arbitrage.

However, the magnitude of savings is contingent upon forecast quality. The Hybrid model yielded the highest aggregated savings (approximately 2.29%), followed closely by the TCN (about 2.02%), whereas the Transformer provided only marginal benefit (about 0.38%). These results imply a causal chain: reduced long-horizon forecast error improves charge/discharge timing (particularly during peak windows), which stabilizes State-of-Charge (SoC) trajectories and increases realized economic benefit. The relatively close performance between the Hybrid and TCN models further indicates that control performance is sensitive to forecast smoothness and bias, not merely average error; thus, reporting both forecasting metrics and operational cost outcomes is essential.

It is important to note that this study treats the scheduling experiment as an interpretable, reproducible benchmark rather than an implementation of an optimal controller. Nevertheless, it directly addresses practical relevance by translating forecast improvements into measurable monetary outcomes.

Challenges and limitations

Several limitations must be acknowledged. (i) *Simplified tariff and battery model*: The simulation employs a fixed TOU tariff and excludes demand charges, export compensation, battery degradation costs, and inverter constraints beyond simple power/SoC limits; consequently, absolute savings are scenario-dependent. (ii) *Heuristic control policy*: The controller is rule-based and forecast-sensitive but non-optimized (e.g., lacking MPC or reinforcement learning). Thus, the reported savings represent a conservative proof-of-concept rather than an upper bound on potential performance. (iii) *Generalizability*: Results are derived from a specific dataset and household context; transfer to other regions may require recalibration due to variations in climate regimes, appliance ownership, and tariff structures. (iv) *Evaluation scope*: Not all baselines were evaluated on both horizons, and inference efficiency metrics depend on the specific hardware and software stack utilized.

Future research directions

Two avenues for future research appear particularly promising. First, *probabilistic forecasting* (e.g., quantile regression, distributional heads, or conformal prediction) could provide calibrated uncertainty intervals, enabling risk-aware dispatch and reserve sizing rather than relying solely on point forecasts. Second, *transfer learning and domain adaptation* could enhance generalization across households, seasons, or regions. Pre-training on large multi-household corpora followed by fine-tuning (or feature-aligned adaptation) would reduce retraining costs and support rapid deployment in data-scarce settings.

From a control perspective, integrating the Hybrid forecasts into *model predictive control* (MPC) or *reinforcement learning* (RL) frameworks would allow for principled benchmarking against optimized scheduling policies. This approach would also address concerns regarding the limited comparison to alternative forecast-driven control strategies. Finally, extending the evaluation to feeder-level aggregation and multi-building

coordination would bridge the gap between household forecasting and utility-scale operational planning.

Conclusion

This study presents an end-to-end, explainable hybrid Transformer-BiLSTM framework for multi-horizon (24,h and 168,h) household energy demand forecasting with weather integration. The model combines self-attention to capture long-range dependencies between demand and weather with bidirectional recurrence to refine local temporal transitions, and it is tuned using Bayesian hyperparameter optimization under a leakage-free chronological evaluation protocol.

Empirically, the proposed hybrid model is competitive at the 24-hour horizon, achieving performance comparable to strong deep-learning baselines (with variations depending on the specific metric), and demonstrates its clearest advantage at the 168-hour horizon, where it attains the best overall accuracy among the evaluated models. Diebold-Mariano tests further confirm that the hybrid model's gains are statistically significant against key baselines, particularly for the weekly horizon.

Interpretability analysis using SHAP indicates that medium-term forecasts are driven primarily by a combination of electrical conditions (e.g., voltage, power factor, apparent/reactive power) and meteorological variables (e.g., pressure and temperature statistics). This yields explanations consistent with physical expectations, thereby supporting operational trust. Robustness experiments show that the model is stable under mild, scale-aware noise and degrades less severely than the comparators under feature dropout, suggesting improved resilience to imperfect sensor and weather inputs.

To assess the operational value of accuracy, we integrated the model forecasts into a forecast-informed household battery scheduling simulation under a fixed time-of-use tariff and consistent battery constraints. Across aggregated multi-week windows, the hybrid forecasts achieved the largest cost reduction relative to the no-battery baseline (approximately 2.29% total cost savings in our setup), demonstrating that improved medium-horizon accuracy translates into measurable economic benefits through optimized charge-discharge timing.

Several limitations should be acknowledged. First, the results are based on a single-site dataset, which limits generalizability across regions, household types, and tariff structures. Second, the scheduling experiment is intentionally a proof-of-concept; the controller is heuristic and does not benchmark against optimized control strategies (e.g., MPC), nor does it account for battery degradation, export compensation, or demand charges. Third, the study emphasizes point forecasting; uncertainty is not explicitly quantified.

Future work will therefore focus on (i) probabilistic forecasting (e.g., quantile or distributional outputs with calibration) to support risk-aware planning, (ii) transfer learning and domain adaptation to improve cross-household and cross-climate generalization, and (iii) tighter integration with optimization-based or learning-based control (MPC/RL) to provide rigorous benchmarks for forecast-driven energy management. Overall, the proposed framework advances transparent, robust, and operationally relevant forecasting for energy informatics applications.

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Author contributions

M.M.S. contributed to conceptualization, methodology, data curation, formal analysis, and preparation of the original draft. A.N. contributed to supervision, validation, writing review, and editing. H.I. contributed to supervision and project administration. All authors read and approved the final manuscript.

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Declarations

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Consent for publication

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Competing interests

The authors declare no competing interests.

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